

Climate Downcasting via Frequency-Domain Autoregressive Transformer Modelling

Goal: Downcasting of climate predictions by earth system models to higher resolution

Setup: Train model on global reanalysis data

Pipeline:

Input: Weather data (here: wind speed) with equiangular resolution of shape (N_{lat}, N_{lon})



Spherical Harmonic transform

Complex coefficients in lower triangular matrix of shape (N_{lat}, N_{lat})



Flattening, element-wise standardization, `torch.view_as_real`

Coefficients of shape $(L, 2)$ where $L = \text{triangular_number}(N_{lat})$



Transformer model that predicts distribution of next coefficient by parametrization as gaussian mixture model.

Autoregressively predict next coefficients starting at L , ending at $T = \text{triangular_number}(2 * N_{lat})$

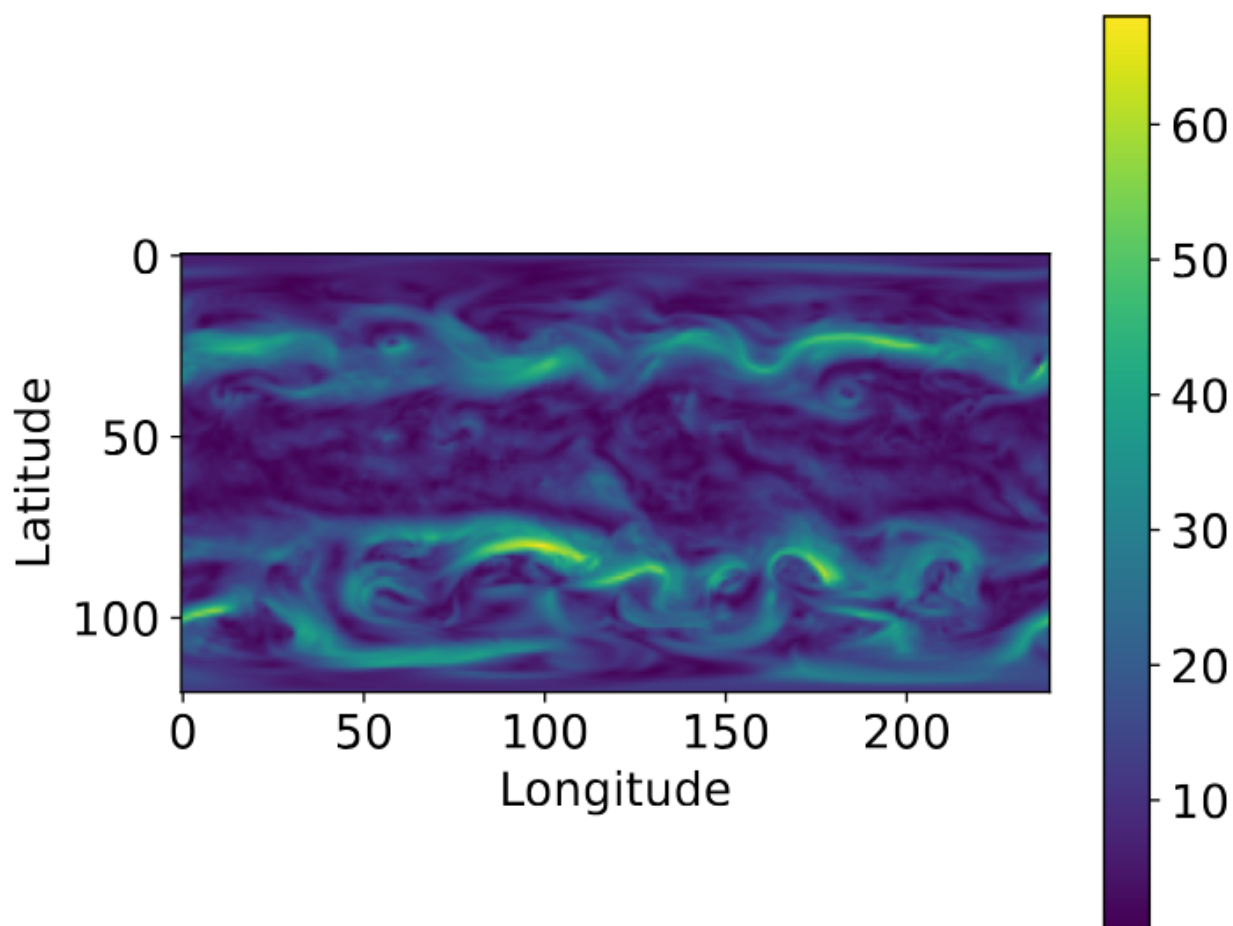
Model output of shape $(T, 2)$



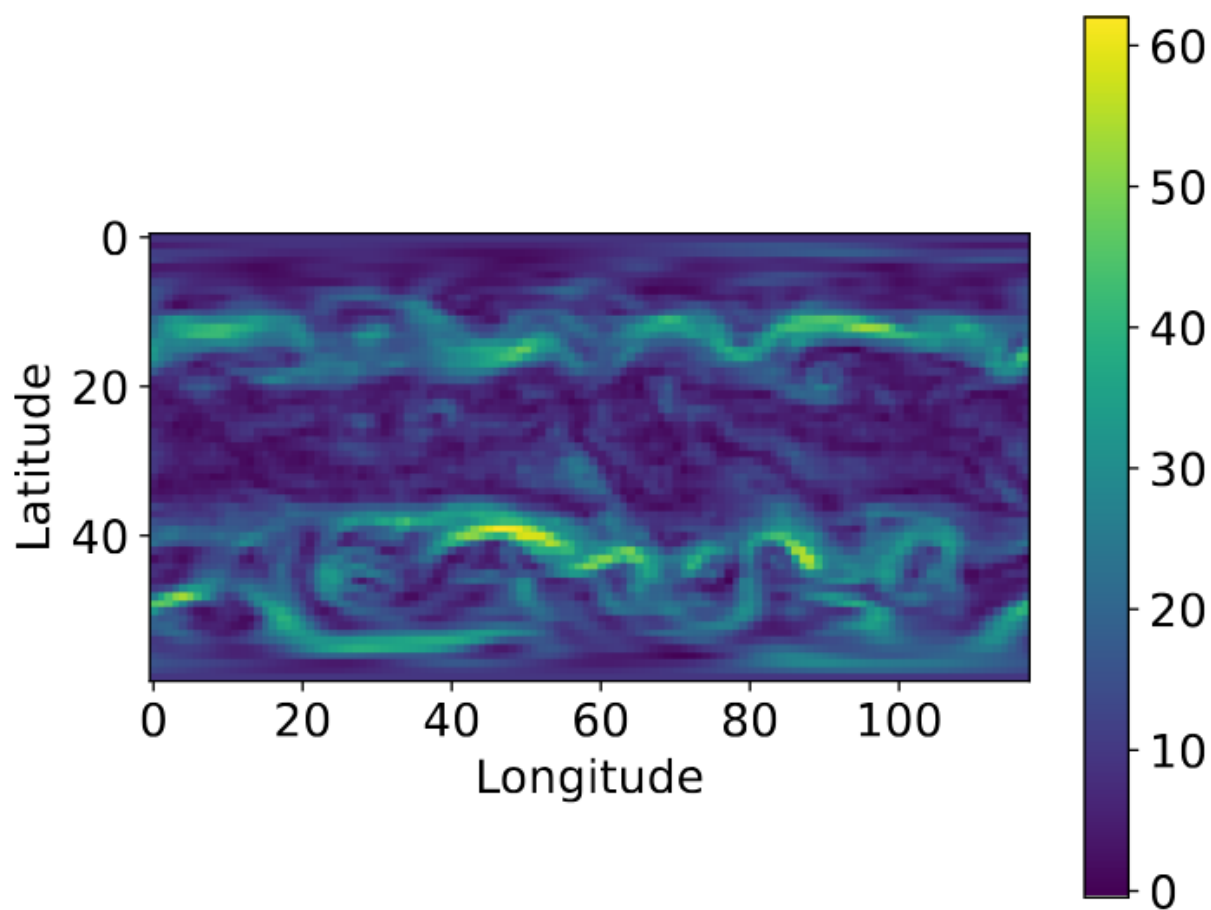
`torch.view_as_complex`, un-standardization, unflattening, inverse SH transform

Model prediction of shape $(2 * N_{lat}, 2 * N_{lon})$

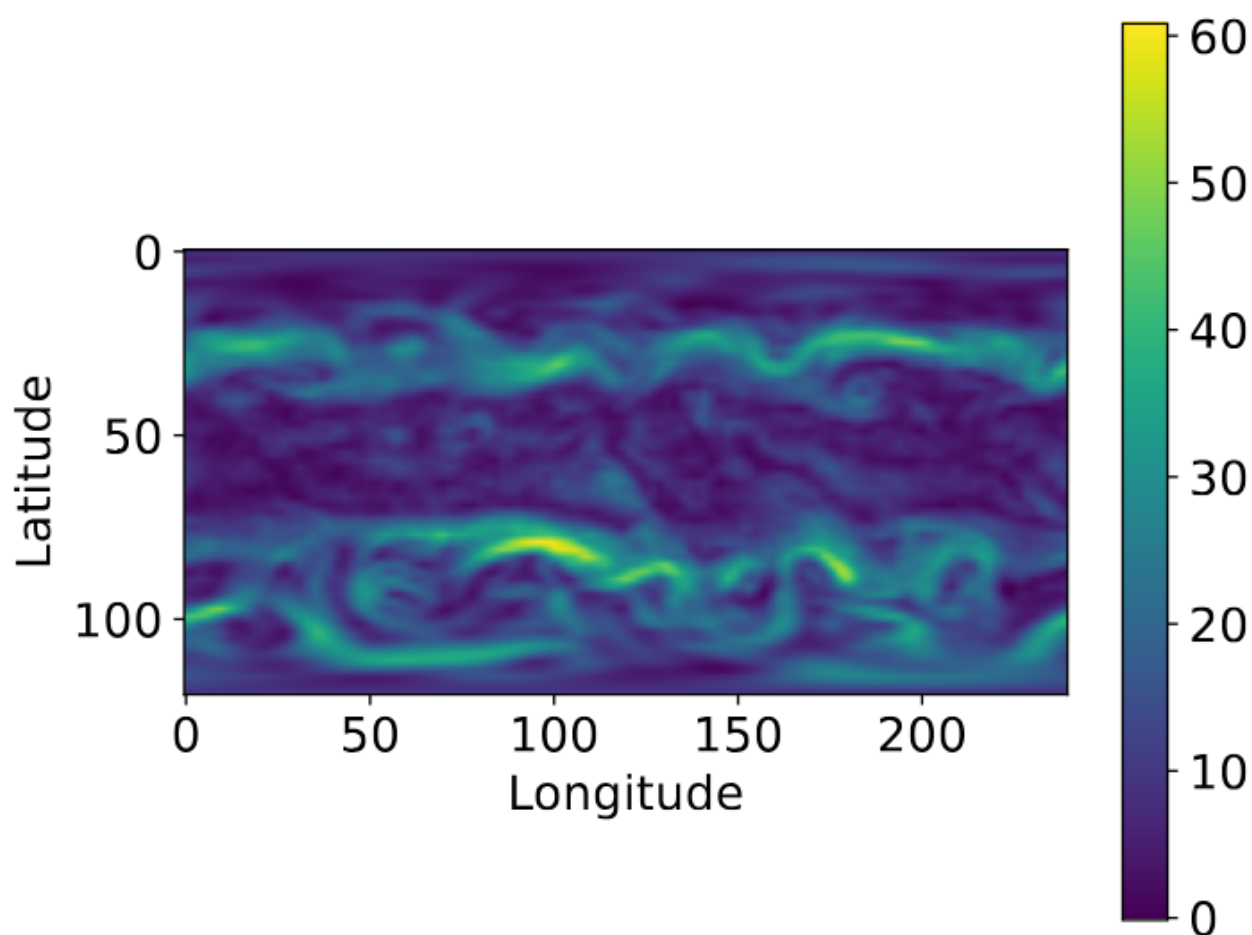
Ground truth data:



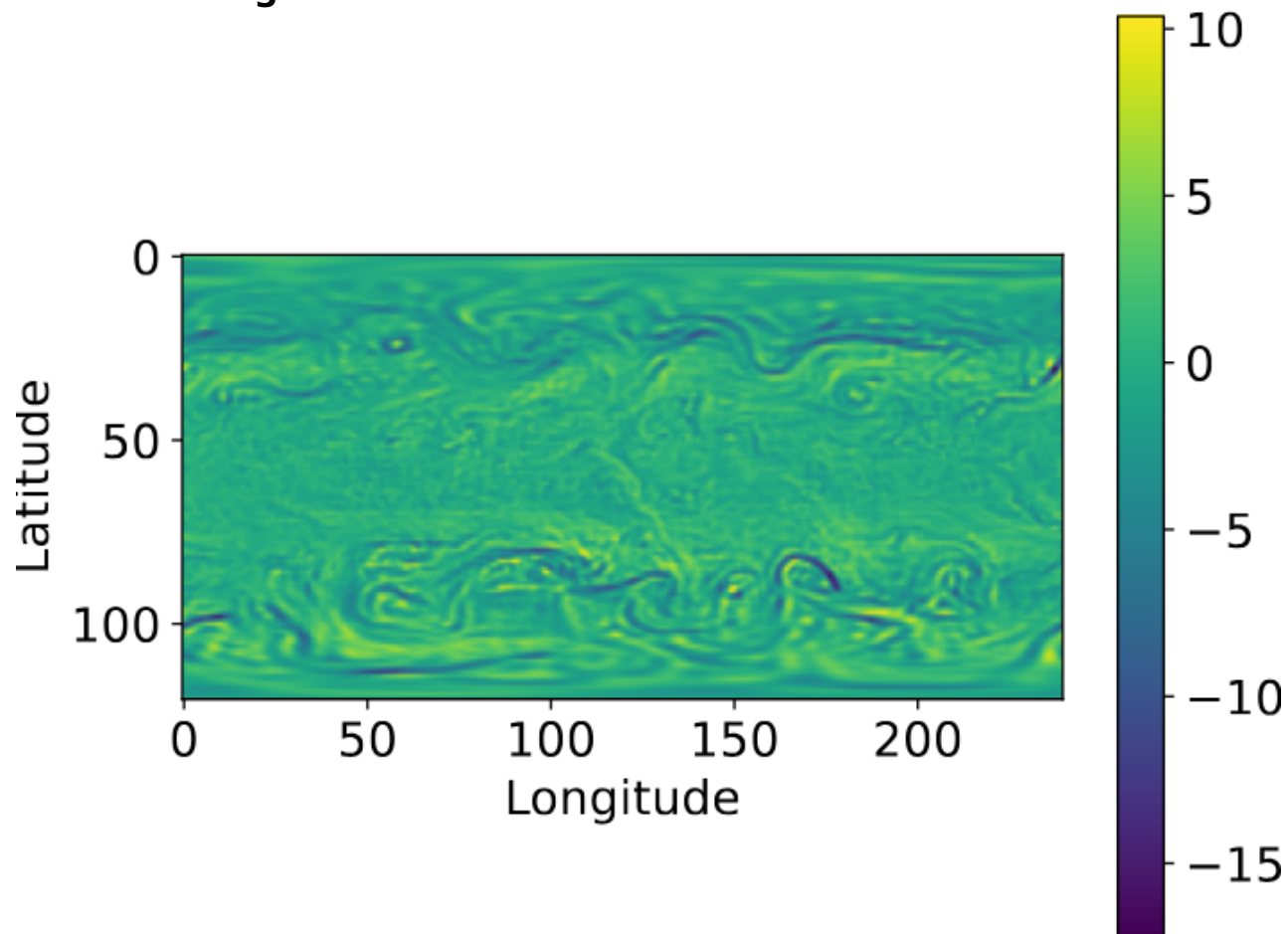
Model input:



Bilinear Interpolation:

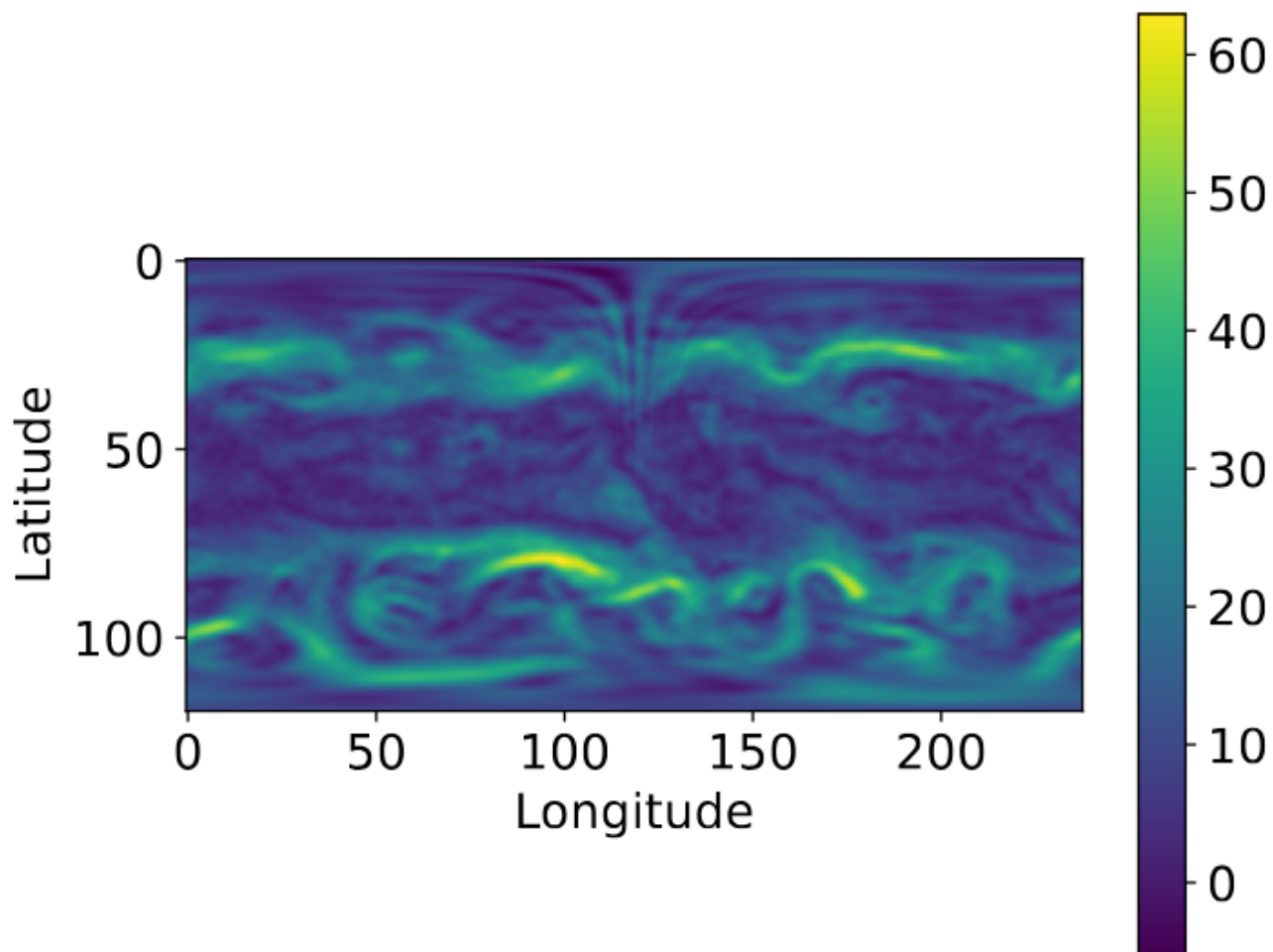


Difference to ground truth:

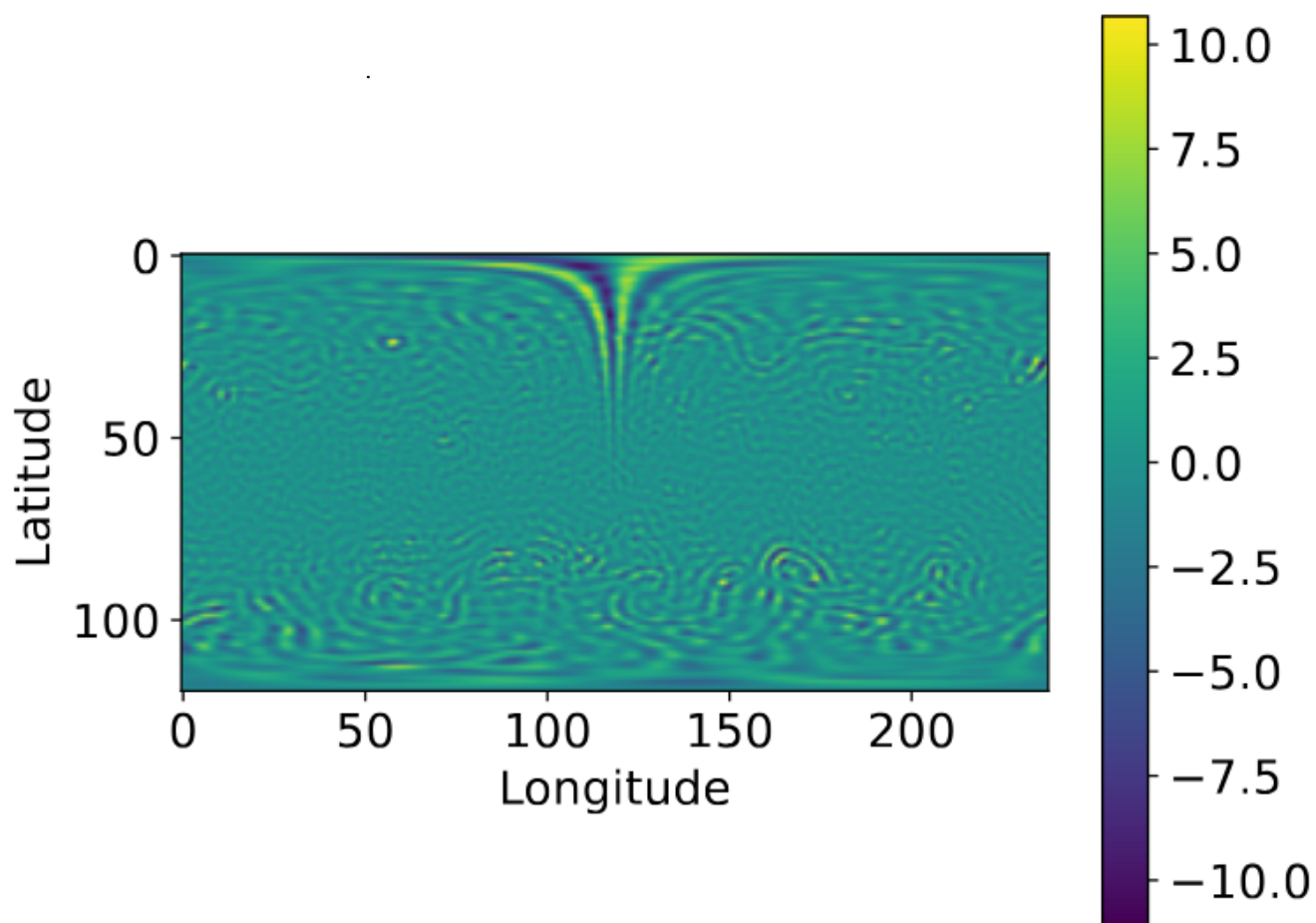


RMSE: 2.12 m/s

Transformer model: 16.3M parameters

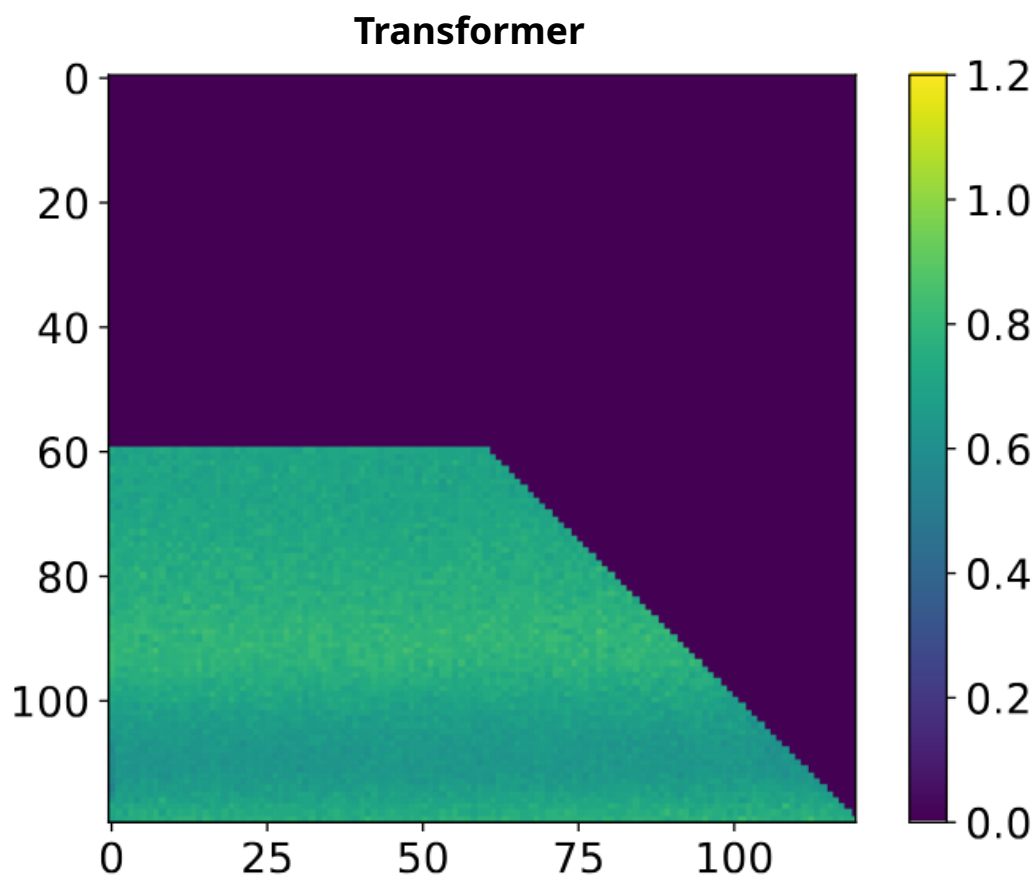
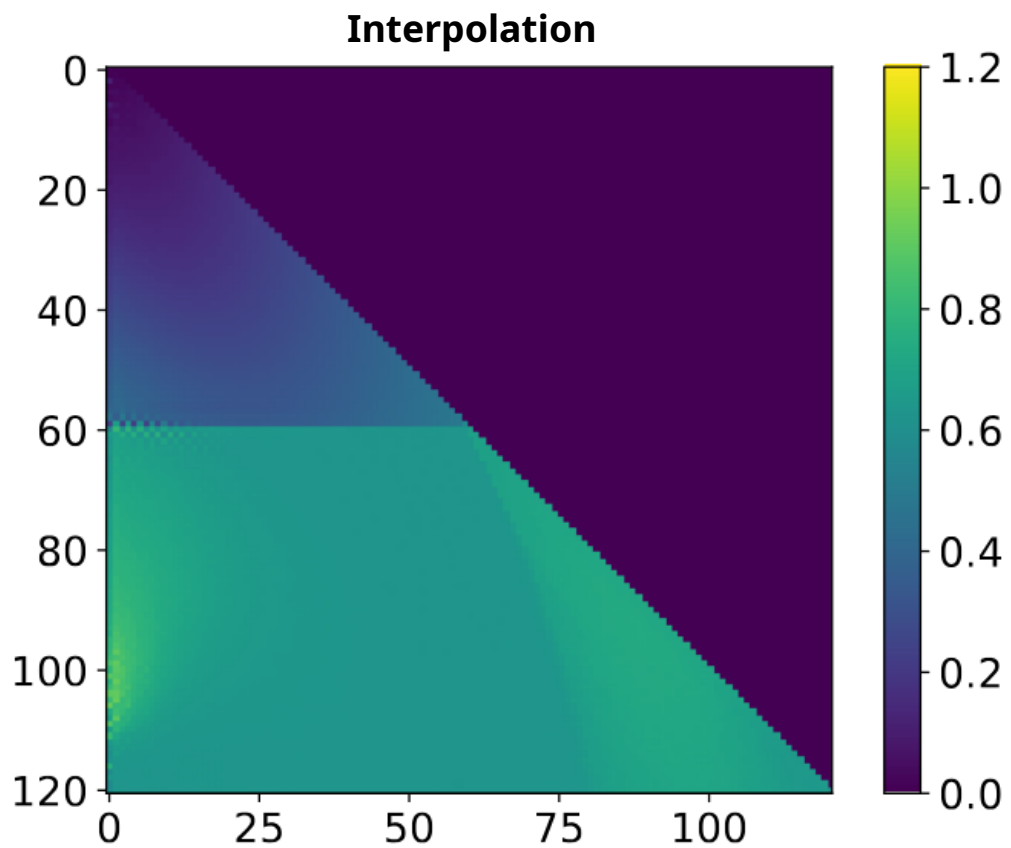


Difference to ground truth:



RMSE: 1.76 m/s

Coefficient RMSE Plots:



**=> Predictions don't seem to get significantly worse
in later autoregression steps**

Next steps:

- Find out where north pole artifact comes from
- Use KV-Caching to make inference faster
- Train larger model